Electric car: launch the design

antocciphysics

UNIT 8 / DAY 1 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A single-loop toy calculation gives 3 V induced emf. Explain why this alone cannot predict the effects on a whole power grid; name two missing features.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: average speed = distance/elapsed time. Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: A design claim becomes testable when it names a measurement. "The car is fast" can become "The car travels 3 m in under 2 s on this level track." That condition corresponds to average speed above 1.5 m/s.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Marzano check - circle: ① ② ③

03 Represent and interpret

Sketch your proposed car. Label the battery, switch, motor, driven axle and wheels. Add a theme that will not obstruct moving parts.



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

04 Solve and justify

A supplied design has a 3 V supply and expected .40 A motor current. Estimate electrical power. Identify at least three reasons this is not all useful forward-motion power.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

Write one measurable design goal and one constraint. Explain how you will measure the goal instead of judging only by appearance.

Marzano check - circle: 1 2 3

06 Choose a method

A circuit has 4 identical series resistors. Add two more at unchanged ideal supply voltage. Compare new current with the original.

Marzano check - circle: 1 2 3

07 Exit check

Create a fair first-test plan naming distance, timing method, number of trials and variables held fixed.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Choose a speed target and a reliability target for a 3 m course. Design repeated trials and a rule for judging success without selecting only the fastest run.

Marzano check - circle: 1 2 3